Parities, differences and neighbours: OEIS A183318

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Abstract

OEIS A183318 carries an empirical recurrence of order 64 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. The entry's condition is decided by a bounded amount of state carried down the array; the admissible windows are the vertices of a finite digraph and the arrays counted are exactly the walks in it. Hence $a(n)$ is a walk count on $S = 64$ vertices and is C-finite, and whether the conjectured recurrence annihilates it is settled by a finite exact computation, with the Cayley–Hamilton theorem bounding the work.

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1 The conjecture

OEIS A183318 is "Number of nX6 binary arrays with an element zero only if there are an even number of ones to its left and an even number of ones above it.". It has offset 1 and begins

$$21, 119, 1109, 7833, 61755, 455509, 3437801, 25528515, \ \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n)=a(n-1)+155*a(n-2)-83*a(n-3)-10499*a(n-4)+2739*a(n-5)+418493*a(n-6)-42935*a(n-7)-11114743*a(n-8)+187159*a(n-9)+210582515*a(n-10)+4942581*a(n-11)-2969847775*a(n-12)-106170955*a(n-13)+32076857369*a(n-14)+1009462397*a(n-15)-270649716999*a(n-16)-4937331775*a(n-17)+1809326210403*a(n-18)+1914644979*a(n-19)-9680756335999*a(n-20)+163670254649*a(n-21)+41751512005591*a(n-22)-1408662580253*a(n-23)-145837266421261*a(n-24)+7057415803719*a(n-25)+413726670171893*a(n-26)-24987326051091*a(n-27)-954307354257023*a(n-28)+66458580117683*a(n-29)+1789004571454345*a(n-30)-136365538707579*a(n-31)-2720716316400381*a(n-32)+218575487621235*a(n-33)+3345872171110277*a(n
> 34)-274993070105947*a(n-35)-3312136548981459*a(n-36)+271453232272799*a(n-37)+2623620146984753
> 38)-209286828757487*a(n-39)-1650724693067001*a(n-40)+124964898554049*a(n-41)+817527401509143*
> 42)-57057707865601*a(n-43)-315201100152543*a(n-44)+19563658984353*a(n-45)+93322287344023*a(n-46)-4906758788153*a(n-47)-20850409285435*a(n-48)+864282086203*a(n-49)+3434976022529*a(n-50)-99411558905*a(n-51)-404036952399*a(n-52)+6269911051*a(n-53)+32355800903*a(n-54)-65276327*a(n-55)-1635520523*a(n-56)-16406321*a(n-57)+45651461*a(n-58)+790773*a(n-59)-534995*a(n-60)-5413*a(n-61)+1739*a(n-62)-a(n-63)-a(n-64)

As of the "Last modified" line on the live entry (Oct 03 2025 01:33:51, revision 9) this is still recorded as empirical, and nothing on the entry records it as proved.

2 The count is a walk count

The entry counts arrays with 6 columns over the alphabet $\{0, 1\}$, growing one row at a time, and asks that an entry may be zero only where the number of ones to its left in its own line and the number of ones above it in its own column are both even.

The count of ones above a cell reaches back to the top of the array, so no window of consecutive rows holds it. It does not have to: only the parity of that count is ever asked for, and a parity is one bit per column, flipped by each one that lands in that column. The other half of the condition, the ones to a cell's left, is read straight off the row being placed. So the state is 6 bits and nothing else, and a row may follow a state exactly when every zero in it sits where both parities are even.

Merging vertices with identical futures leaves $S = 64$. An admissible array is exactly a walk, so with M the adjacency matrix and ι, τ the vectors recording which states may begin and end an array,

$$a(n) \;=\; \iota^\top M^{n-1}\tau,$$

the exponent fixed by the entry's own shape and checked against its published terms. In particular a is C-finite.

3 The proof

Lemma 1. *Let $q(t) = t^{64} - \sum_i c_i t^{64-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M)\tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^\top M^j q(M)\tau = 0$ for all large j, and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+64} - \sum_i c_i M^{j+64-i}$. For the bound, M^S is an integer combination of $I, M, \ldots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. A constant factor in front of a divides out of the residual and changes nothing here. □

Theorem 2.

$$a(n) \;=\; a(n-1)+155\,a(n-2)-83\,a(n-3)-10499\,a(n-4)+2739\,a(n-5)+418493\,a(n-6)-42935\,a(n-7)-111147$$

for every $n > 64$.

Proof. The vector $w = q(M)\tau$ was formed by 64 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \ldots,$ again exactly; those integers vanish from the index corresponding to $n = 64 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. □

4 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 19 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the digraph is built by reading the entry's English, and a misreading gives different counts.

Second, the reading was pinned before the digraph was written, by a program that enumerates every array of the given shape one by one and tests the entry's condition on it directly. That program shares no code with the transfer matrix, so an error in one does not hide an error in the other.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, `https://oeis.org`, 2026. Sequence A183318.

[2] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)

[3] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009.

[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.